



DEEP LEARNING PROJECT

# Hotel Review Analyzer MLP Project

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# Why analyze hotel reviews?

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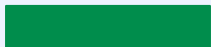
Reading reviews one by one takes time.

## Many individual reviews

“房间很干净，但晚上很吵。”  
“地铁很近，前台态度一般。”  
“床很舒服，隔音需要改善。”



## One visual summary

|                  |                                                                                     |          |
|------------------|-------------------------------------------------------------------------------------|----------|
| Cleanliness      |  | Positive |
| Sound insulation |  | Negative |
| Location         |  | Positive |

Extract aspects + sentiment to compare strengths and recurring problems.

A visual summary helps people make hotel-booking decisions faster.

# The project

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**A user signs in and pastes a hotel review.**

Your application identifies each aspect and its sentiment, then displays review-level scores with supporting sentences.

房间很干净，但是前台态度不好。

| Cleanliness | Service  | Other aspects |
|-------------|----------|---------------|
| Positive    | Negative | Not mentioned |

**The trainable classifier must be an MLP. Each student submits a working website.**

# The complete workflow

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**01**

**Explore the data**

Understand embeddings and inspect similar sentences.

**02**

**Create labels**

Build an annotation tool and verify suggested labels.

**03**

**Train an MLP**

Compare architectures and evaluate unseen examples.

**04**

**Deploy the app**

Sign in, analyze a review and show aspect scores.

**AI coding is allowed. You are responsible for the data, experiments and predictions.**

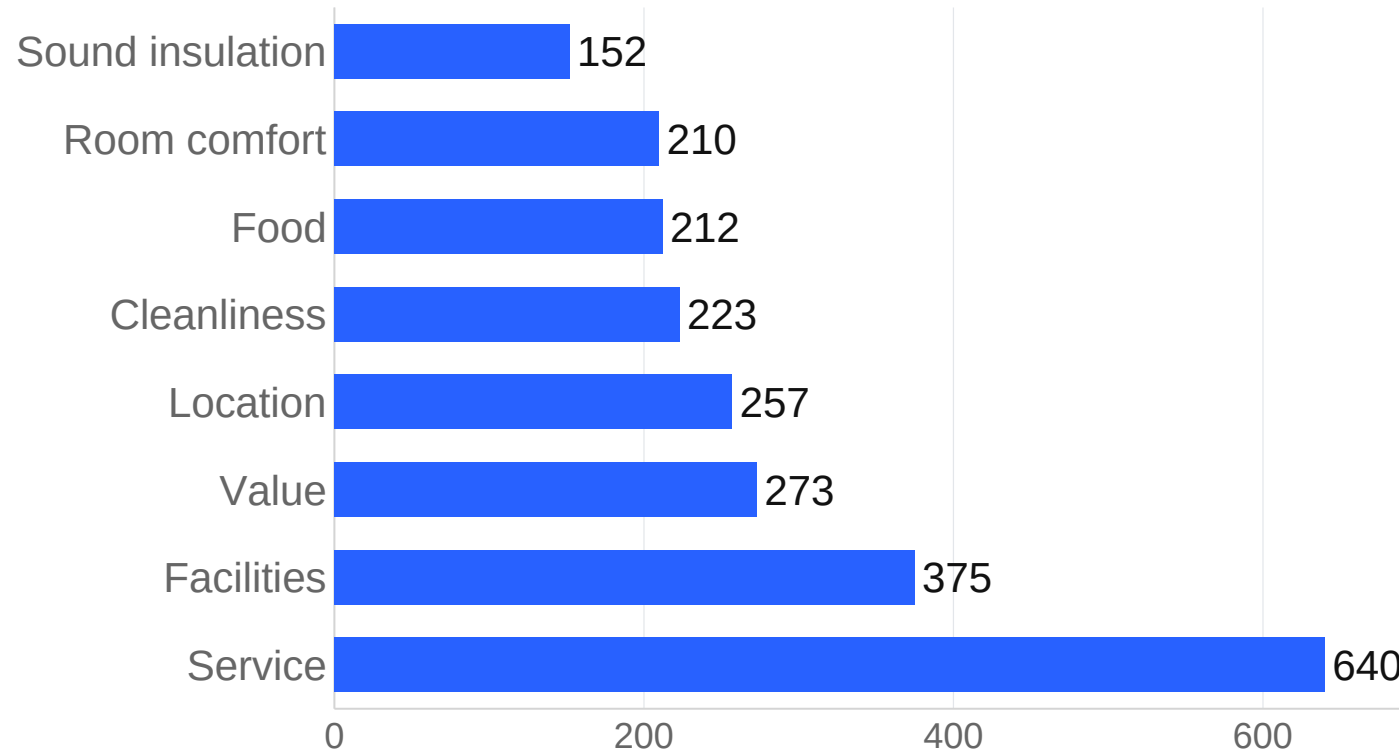
# The provided dataset

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| Resource                | What you receive                                  |
|-------------------------|---------------------------------------------------|
| Chinese hotel sentences | 2,000 sentences from 2,000 source reviews         |
| Sentence embeddings     | 2,000 × 256 float32 values                        |
| Identifiers and context | sentence_id, review_id and adjacent sentences     |
| Labels                  | Unlabeled: you create aspect and sentiment labels |
| Source                  | Public historical hotel-review corpus             |

The CSV and embedding rows match. Join later annotations by sentence\_id.

## What appears in these sentences?



**Noise and soundproofing deserve a separate aspect.**

A sentence may mention several aspects.

Keyword hits are clues. They are not final labels.

**Exploratory keyword counts over all 2,000 sentences, before annotation review.**

# Eight proposed aspects

| Index | Aspect                   | Include                                       |
|-------|--------------------------|-----------------------------------------------|
| 0     | Cleanliness              | Dirt, hygiene, stains and pests               |
| 1     | Service                  | Staff behavior, help and check-in handling    |
| 2     | Location                 | Access, transport and nearby places           |
| 3     | Facilities               | Equipment, amenities and condition            |
| 4     | Room comfort             | Space, bed comfort and temperature            |
| 5     | Sound insulation & noise | Soundproofing, hallway / traffic / room noise |
| 6     | Food                     | Hotel breakfast and dining quality            |
| 7     | Value                    | Prices, charges and value for money           |

Use these aspect names and this order across labels, model outputs and the website.

# Aspect boundaries

| Text example | Aspect           | Reason                   |
|--------------|------------------|--------------------------|
| 空调坏了         | Facilities       | Equipment does not work  |
| 空调太吵，整晚睡不好   | Sound insulation | Noise affects sleep      |
| 床单很脏         | Cleanliness      | Hygiene                  |
| 酒店早餐很贵       | Value            | Price of hotel breakfast |
| 酒店早餐很难吃      | Food             | Food quality             |
| 附近有很多餐厅      | Location         | Access to nearby dining  |

**Label the property discussed, not every object word. Add aspects for distinct claims.**

# EMBEDDINGS

# What is a sentence embedding?

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服务很好，一直住这个酒店。

A pretrained encoder represents the sentence with 256 numbers.

$$\mathbf{x} = \begin{bmatrix} x_0 & x_1 & x_2 & \dots & x_{255} \end{bmatrix}^T$$

Related sentences can have similar vectors.

The MLP learns how these vectors relate to our labels.

**The 256 dimensions do not correspond to 256 named topics.**

## Similar vectors, similar sentences

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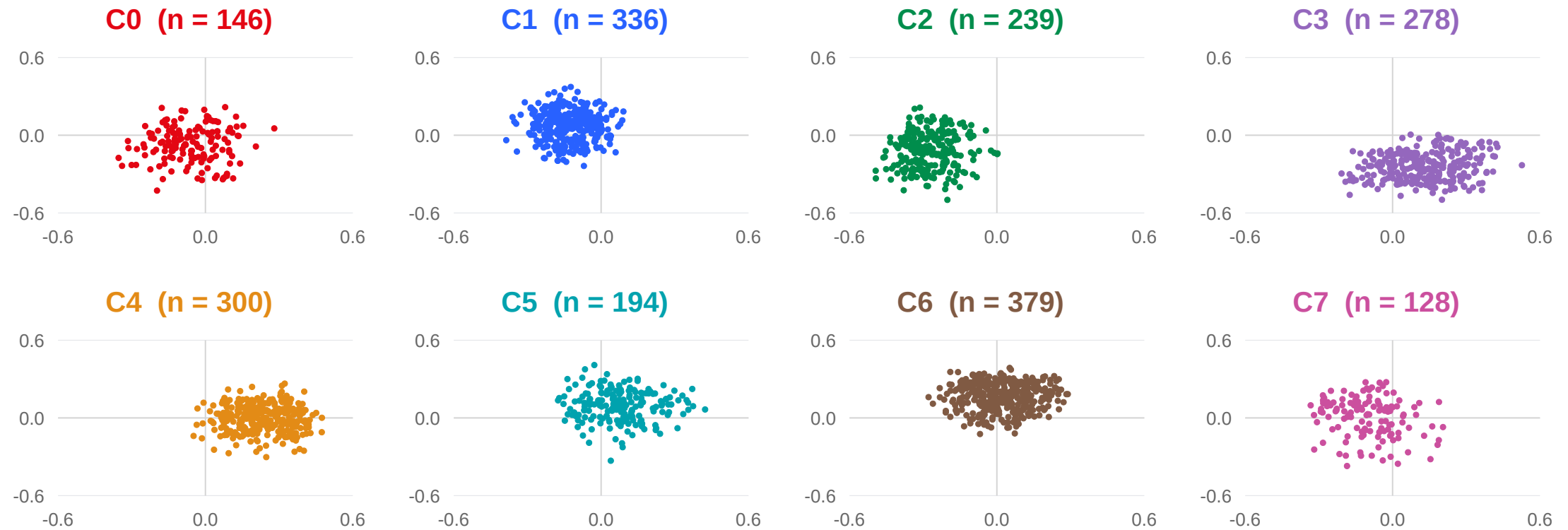
Query: HTL\_S0980

服务很好，一直住这个酒店。

| Nearby sentence         | Cosine similarity |
|-------------------------|-------------------|
| 每次出差都住在这家酒店，酒店在服务上做得很好。 | 0.877             |
| 第二次入住，酒店的服务还是一如既往地好！    | 0.843             |
| 对酒店的服务非常满意…提供的时间承诺都能遵守… | 0.802             |

**Similarity helps exploration. It does not guarantee the same aspect or sentiment.**

## A 2D view of eight clusters



Same PCA axes in every panel. Variance shown: 11.8%. Cluster IDs are not aspect labels.

# What do the clusters contain?

| Cluster | Sentences | Themes in inspected examples                  |
|---------|-----------|-----------------------------------------------|
| C0      | 146       | Noise, poor sound insulation and sleep        |
| C1      | 336       | Positive room and value evaluations           |
| C2      | 239       | Location, transport and surroundings          |
| C3      | 278       | Mixed narratives, complaints and some replies |
| C4      | 300       | Booking, check-in and billing problems        |
| C5      | 194       | Service praise and repeat stays               |
| C6      | 379       | Room condition, cleanliness and complaints    |
| C7      | 128       | Breakfast and hotel dining                    |

**Service appears in several clusters. One cluster can contain several aspects.**

# Your embedding exploration

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- Load** Check X.shape and match several sentence IDs to the CSV.
- Cluster** Run k-means. Inspect 5 nearby sentences in each cluster.
- Compare** Try another k. Do the groups change?
- Explain** Show one useful grouping and one misleading grouping.

**Use the full 256-dimensional vectors for clustering and MLP training.**

# DATA ANNOTATION

# Data quality comes before labels

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Some records contain hotel replies rather than guest opinions.

“给您带来不便我们深表歉意！”

27 records match possible hotel-reply phrases. Review each one.

| Record decision | What to do                                         |
|-----------------|----------------------------------------------------|
| Keep            | Readable guest opinion or description              |
| Needs review    | Ambiguous wording or uncertain sentence boundary   |
| Exclude         | Hotel reply, mixed reply text or unusable fragment |

**Keep sentence\_id.** If you edit the text, its old embedding no longer matches.

# One state for each aspect

| Code | State    | Meaning                                            |
|------|----------|----------------------------------------------------|
| 0    | Absent   | The sentence does not mention this aspect          |
| 1    | Negative | An unfavorable evaluation                          |
| 2    | Neutral  | Mentioned, with no clear positive or negative view |
| 3    | Positive | A favorable evaluation                             |
| 4    | Mixed    | Both positive and negative views of this aspect    |

**Eight aspects give eight state labels for each reviewed sentence.**

**Unlabeled and excluded records are workflow states, not prediction classes.**

## A labeled example

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房间里还算干净，就是噪音太大，价格相对来说还是贵的

| Aspect             | State        | Evidence            |
|--------------------|--------------|---------------------|
| Cleanliness        | Positive (3) | 还算干净                |
| Sound insulation   | Negative (1) | 噪音太大                |
| Value              | Negative (1) | 价格…还是贵的             |
| Other five aspects | Absent (0)   | No explicit mention |

$$y = [3 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1]$$

# Consistent rules for difficult cases

| Case                       | Rule                                               |
|----------------------------|----------------------------------------------------|
| “早餐一般”                     | Food = neutral under this project convention       |
| “早餐很差”                     | Food = negative                                    |
| “前台热情，但办理太慢”               | Service = mixed within one sentence                |
| “位置很好，但服务很差”               | Location = positive; Service = negative            |
| “以后不会再住了”                  | All aspects absent if no specific aspect is stated |
| Unclear or incomplete text | Mark needs_review; resolve or exclude              |

**Use exact text as evidence. Do not infer missing aspects from an overall rating.**

# A web tool for faster annotation

Hotel annotation

HTL\_S1826    Progress: 20 reviewed

房间里还算干净，就是噪音太大，价格相对来说还是贵的

| Aspect           | Absent   | Negative | Neutral | Positive | Mixed |
|------------------|----------|----------|---------|----------|-------|
| Cleanliness      |          |          |         | Selected |       |
| Service          | Selected |          |         |          |       |
| Location         | Selected |          |         |          |       |
| Facilities       | Selected |          |         |          |       |
| Room comfort     | Selected |          |         |          |       |
| Sound insulation |          | Selected |         |          |       |
| Food             | Selected |          |         |          |       |
| Value            |          | Selected |         |          |       |

Save and next

Needs review

Exclude

Undo / previous

Autosave, keyboard shortcuts, context toggle and CSV / JSON export

# LLM suggestions with human review

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**01**

## Define the prompt

Provide aspect definitions, state codes and a few examples.

**02**

## Request JSON

Return sentence\_id, 8 states and exact evidence spans.

**03**

## Check the result

Validate the schema and let a human correct every record.

**04**

## Save separately

Keep AI suggestions, human labels and review status.

**A suggested label becomes training data only after a human confirms it.**

# A reviewed dataset for training

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## Pilot first

Two people label the same 20 sentences independently.  
Discuss disagreements and update the codebook.

## Review all labels

Check every AI suggestion. Resolve needs\_review records.

## Record exclusions

Keep an exclusion reason and the original sentence\_id.

## Freeze a split

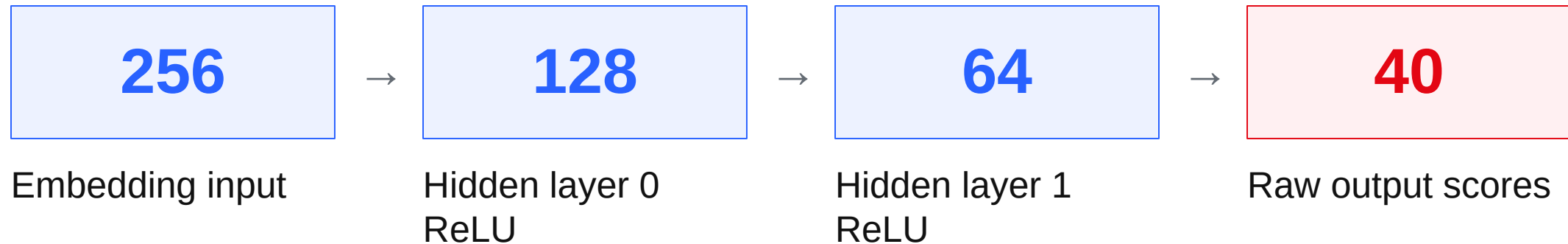
Use reviewed records only. Keep review IDs separate across train, validation and test sets.

**Freeze the label rules and test set before comparing MLP architectures.**

# MLP TRAINING

## An MLP with eight classification outputs

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**Reshape 40 scores into 8 aspects × 5 states.**

Each aspect chooses one state. Different aspects can be present together.

**Use one softmax per aspect, not one softmax over all 40 outputs.**

# Architecture experiments

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| Run | Hidden widths | What you compare           |
|-----|---------------|----------------------------|
| A   | (64)          | A compact starting model   |
| B   | (128, 64)     | More capacity and depth    |
| C   | (128, 64, 32) | An additional hidden layer |

Keep the same labels and data split.

Record learning rate, batch size, dropout, seed and parameter count.

**The learned classifier must be an MLP. LLMs may assist labeling, not replace inference.**

# Training with the reviewed labels

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```
logits = model(X_batch).reshape(-1, 8, 5)
loss = criterion(
    logits.reshape(-1, 5), y_batch.reshape(-1)
)
optimizer.zero_grad()
loss.backward()
optimizer.step()
```

```
criterion = nn.CrossEntropyLoss()
y_batch contains 8 integer state codes per sentence.
```

**Plot training and validation loss. Select the checkpoint using validation performance.**

# Evaluation beyond overall accuracy

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| Question                            | Evidence to report                              |
|-------------------------------------|-------------------------------------------------|
| Did we identify the aspects?        | Mention detection: precision, recall and F1     |
| Did we identify aspect + sentiment? | End-to-end micro-F1 and macro-F1                |
| Which labels are difficult?         | Per-aspect results and class support            |
| Does the model overfit?             | Training / validation loss and metric curves    |
| Why does it fail?                   | At least 5 errors with text and expected labels |

**Report results on the same held-out test set after model selection.**

**Predicting “absent” often can produce high accuracy while missing useful opinions.**

# WEB DEPLOYMENT

# The application students will deliver

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- Sign in** A working login, logout and a test account for the teacher.
- Paste a review** Accept Chinese text with one or several sentences.
- Analyze** Split text, embed sentences and run the saved MLP.
- Explain** Show each aspect score, mention count and evidence.

**Each student submits a URL that the teacher can open and test.**

## Sentence labels become review scores

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| Sentence state   | Value |
|------------------|-------|
| Positive         | +1    |
| Negative         | -1    |
| Neutral or mixed | 0     |
| Absent           | Skip  |

$$S_a = 100 \times \frac{\sum_{i \in M_a} S_{ia}}{n_a}$$

$M_a$  = sentences mentioning aspect  $a$   
 $n_a$  = number of those sentences  
 $S_{ia}$  = sentence contribution

**Range: -100 to +100. If  $n_a = 0$ , show “Not mentioned”.**

**This score summarizes predicted sentiment. It is not model confidence.**

## A worked example: service

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| Sentence           | Service state | Contribution |
|--------------------|---------------|--------------|
| 前台态度不好。            | Negative      | -1           |
| 第二天服务员主动帮我换房，服务很好。 | Positive      | +1           |
| 退房办理得很快。           | Positive      | +1           |

$$\text{Service score} = 100 \times \frac{-1 + 1 + 1}{3} = +33.3$$

Display “3 mentions: 2 positive, 1 negative” next to the bar.

# What the website should show

## Hotel Review Analyzer

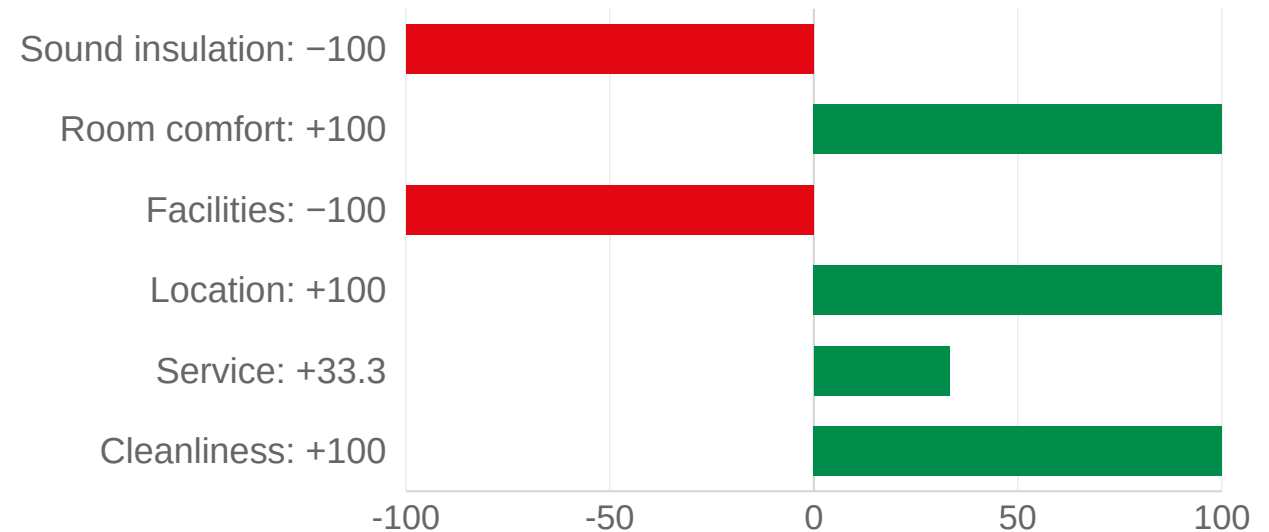
Signed in: student01   Log out

### Paste a review

房间很干净，床也很舒服。  
但是晚上隔音不好。  
空调也不好。  
前台态度不好。  
第二天服务员主动帮我换房，服务很好。  
退房办理得很快。  
去地铁站很方便。

Analyze review

### Review sentiment scores



Food: Not mentioned   Value: Not mentioned

**Service: 3 mentions, 2 positive and 1 negative. [Click to view the sentences.](#)**

## Zero and missing scores mean different things

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| Evidence for one aspect       | Score         | What the user should see       |
|-------------------------------|---------------|--------------------------------|
| No sentence mentions it       | Not mentioned | No bar and zero mentions       |
| One neutral sentence          | 0             | 1 neutral mention              |
| One mixed sentence            | 0             | 1 mixed mention                |
| One positive and one negative | 0             | 2 mentions with opposing views |

**Show state counts and evidence, so the same average does not hide different stories.**

# The same processing at training and deployment

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| Stage              | Required behavior                                 |
|--------------------|---------------------------------------------------|
| Sentence splitting | Use a consistent rule and preserve sentence order |
| Embedding          | qwen3.7-text-embedding, dimensions = 256          |
| MLP                | Load the chosen checkpoint and label order        |
| Prediction         | Choose one of five states for every aspect        |
| Aggregation        | Average only sentences mentioning that aspect     |
| Display            | Show scores, counts and supporting text           |

Keep API keys in the backend. Use `model.eval()` during inference.

## Two-week milestones and final submission

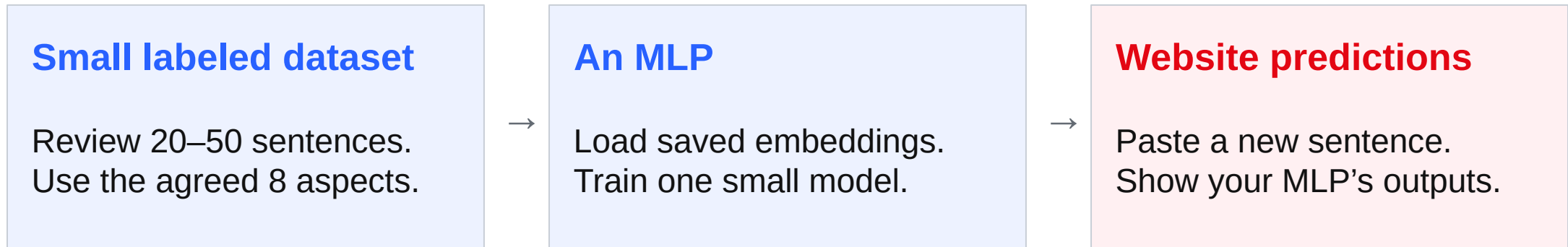
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| Milestone                   | Deliverable                                                   |
|-----------------------------|---------------------------------------------------------------|
| Week 1: data and annotation | Working annotation tool, agreed rules and reviewed labels     |
| Week 1: first pipeline      | Small reviewed dataset → MLP → website predictions            |
| Week 2: experiments         | At least 3 MLP configurations and an error analysis           |
| Week 2: deployment          | Working login, review analysis and evidence display           |
| Final submission            | Website URL, test account, code, labels and experiment report |

**Each student explains one design choice, one failure and one improvement.**

# Tomorrow's minimum goal

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| Time      | Suggested focus                                     |
|-----------|-----------------------------------------------------|
| 0–15 min  | Inspect examples and agree on labels                |
| 15–35 min | Label a small batch and export it                   |
| 35–60 min | Train an MLP using the provided embeddings          |
| 60–85 min | Connect the saved model to a simple prediction page |
| 85–90 min | Paste new text and demonstrate the complete path    |

**First make the complete pipeline work. Expand labels and evaluate it over two weeks.**